PROJECT 18

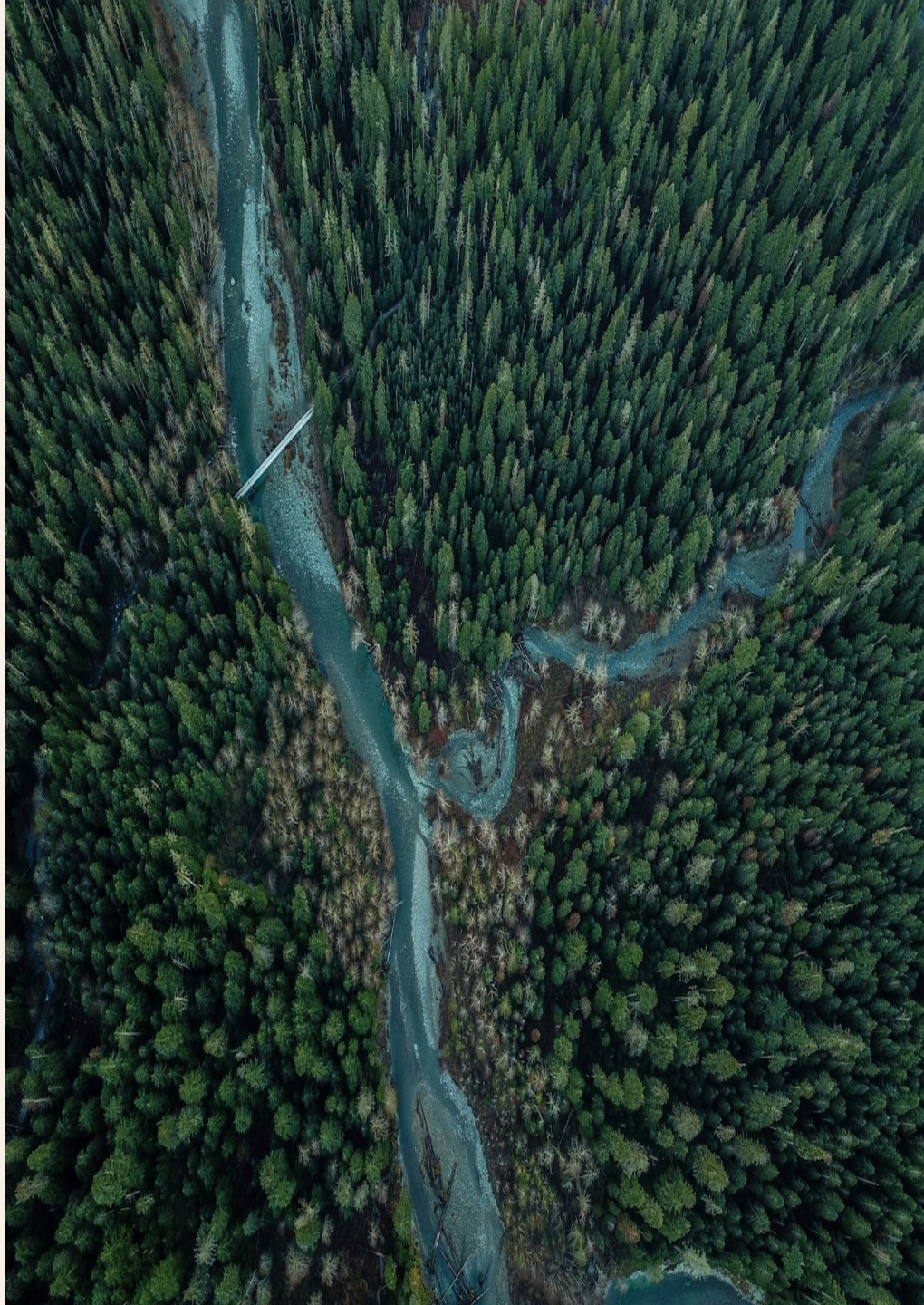
Water Quality Prediction

Gradient boosting and deep neural networks

A chronological study of a calculated surface-water index

TEAM-ID (replace) / SECTION (replace)

Team details are editable dummy values



Problem and research question

Can a small neural network improve on a strong tree model when estimating a documented water-quality index?

What the project delivers

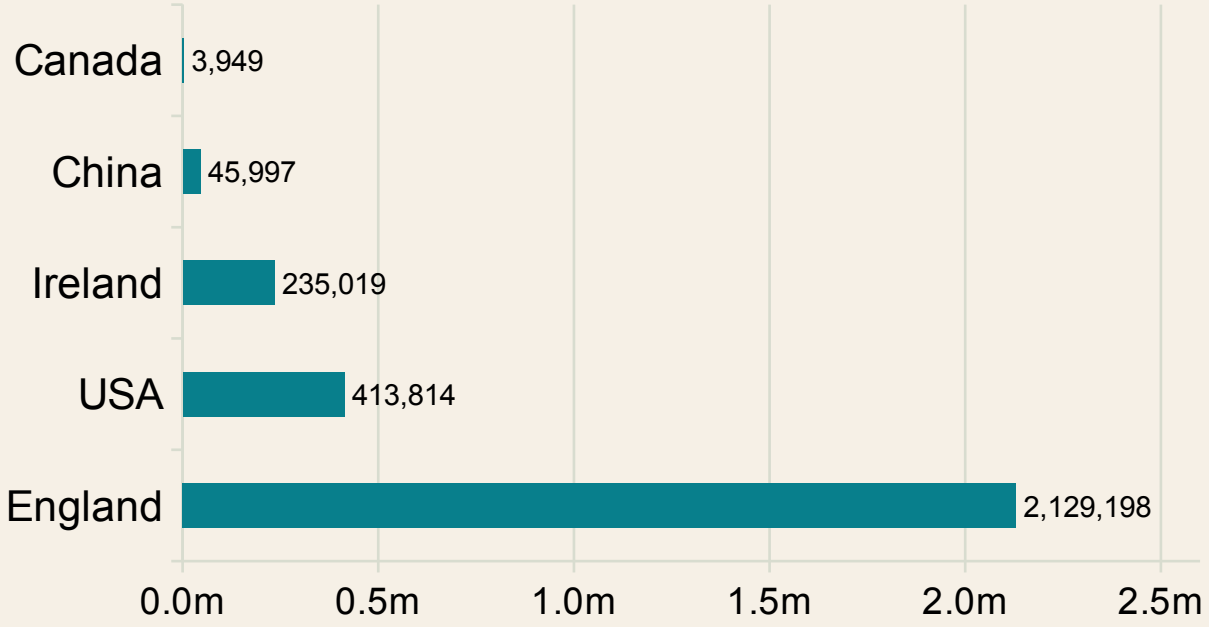
One continuous target on a 0–100 scale
A shared, chronological evaluation protocol
Model comparison and feature explanations
An explicit audit of how the target was created

Prediction inputs must be meaningful for the intended use.

The dataset and eight measurements

2,827,977 rows

1940–2023 · Five countries



Available inputs

Ammonia and BOD
Dissolved oxygen
Orthophosphate
pH and temperature
Nitrogen and nitrate

No turbidity, conductivity or dissolved-solids columns are supplied.

The derived-target audit

100% of released labels

reconstruct within 0.0000001 from the same eight inputs

Source-defined row-level calculation

Upper-only objectives
F1 = F2 for an individual row
Denominator = 12 for eight variables

This evaluates an index emulator.

Official CCME WQI requires a defined assessment period and suitable objectives.

Preprocessing and temporal evaluation

Partition	Period	Available rows	Role
Training	1940–2018	2,478,761	300,000 shared rows
Validation	2019	87,177	Tuning and model selection
Calibration	2020	49,182	Residual interval radius
Testing	2021–2023	202,886	Final frozen comparison

Remove 9,971 exact duplicates. Preserve pollution extremes.

Flag invalid readings. Fit median imputation on training only.
Use asinh and standard scaling for the MLP and Ridge.

Model design and bounded tuning

XGBoost

Thresholds and tabular interactions

Depths 4 and 6
Learning rate 0.05
Up to 1,000 boosting rounds
Validation early stopping

PyTorch MLP

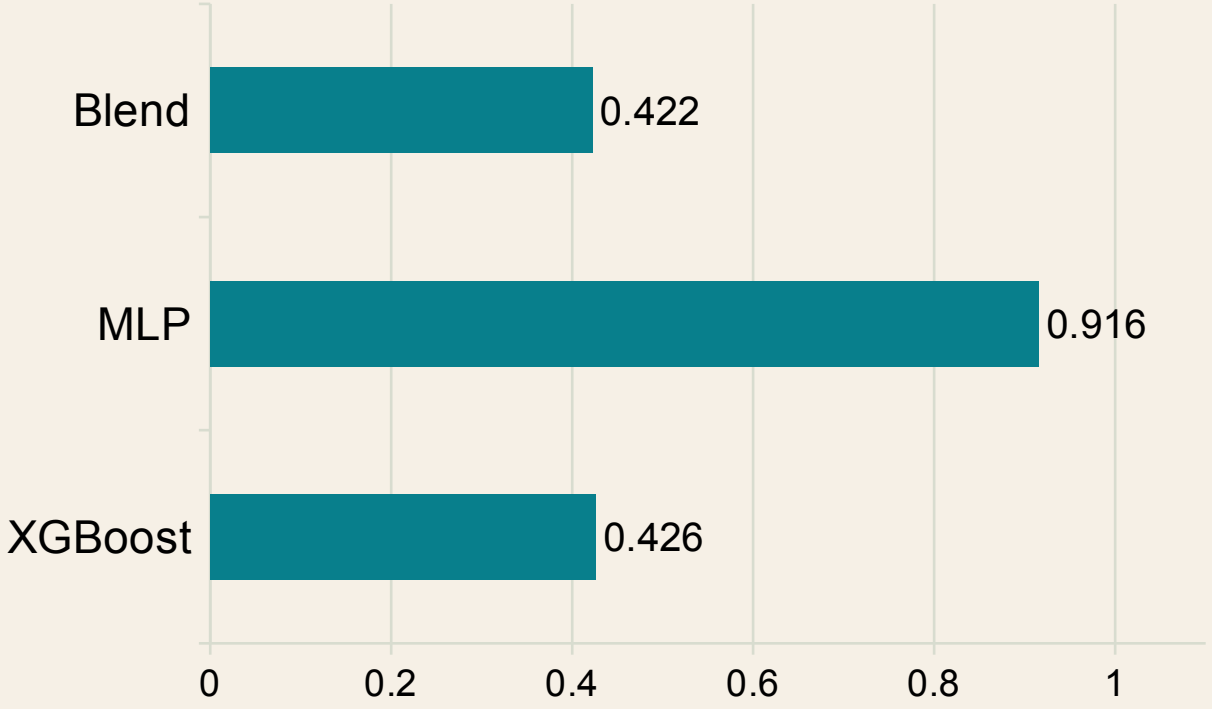
Learned nonlinear combinations

Hidden layers: 64/32 or 128/64
ReLU and dropout 0.1
AdamW with weight decay
Up to 60 training epochs

Both use eight measurements plus eight missing/invalid flags.

XGBoost leads the individual models

Test RMSE in index points

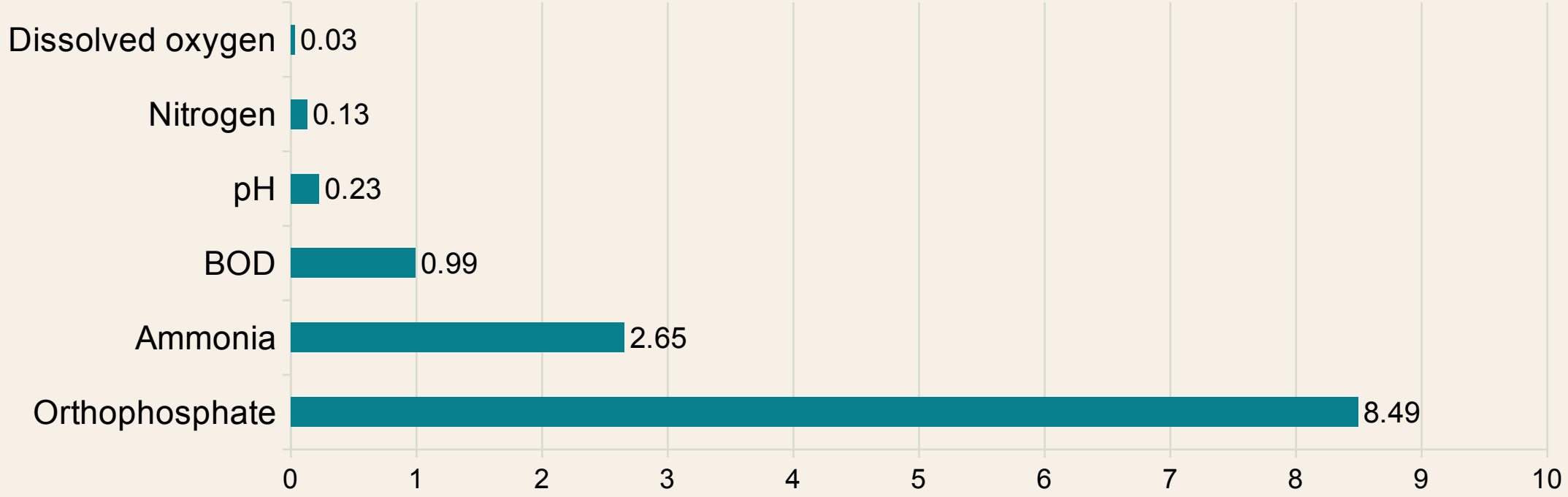


Model	MAE	RMSE	R^2
Mean baseline	11.791	13.458	-0.4141
Ridge	4.344	5.568	0.7580
XGBoost	0.110	0.426	0.9986
MLP	0.447	0.916	0.9934
Blend	0.127	0.422	0.9986

Selected blend
90% XGBoost + 10% MLP

Orthophosphate drives model reliance

Increase in validation MAE after shuffling

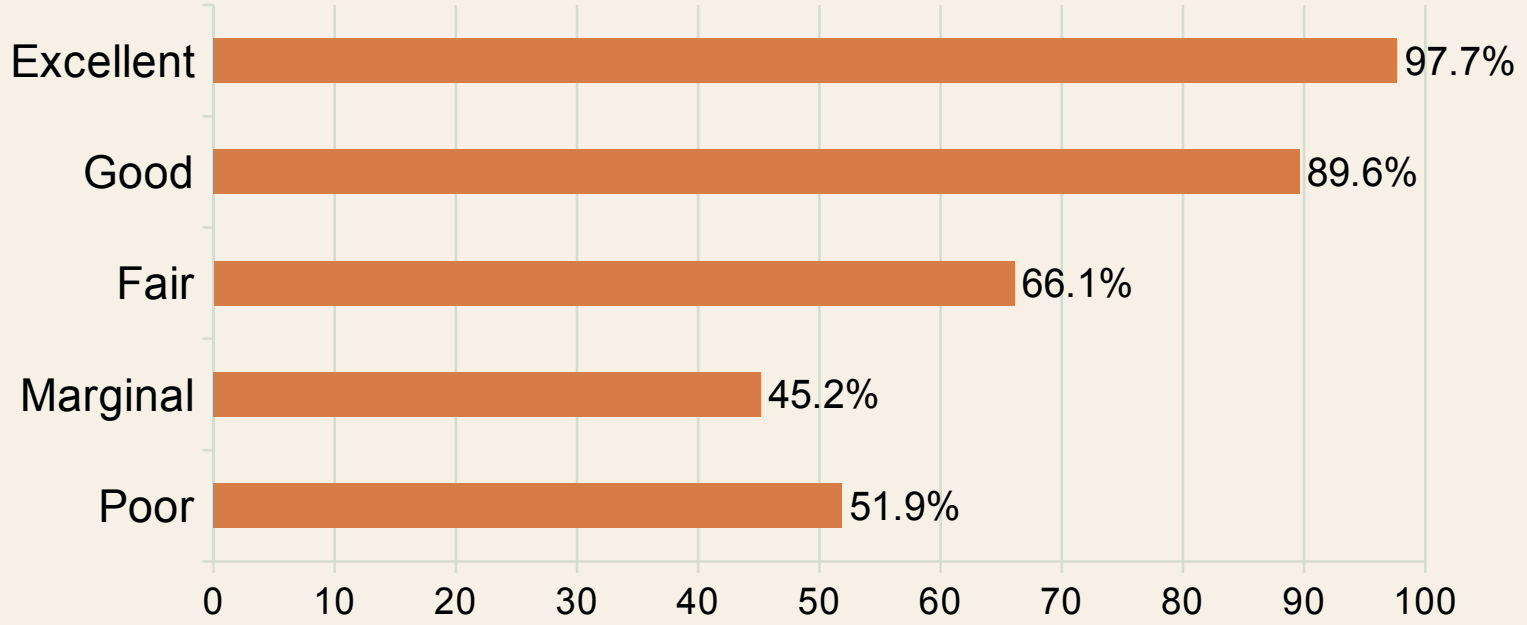


These inputs directly enter the source formula. Attribution does not establish causation.

Interval coverage varies by index band

89.3% overall coverage

Nominal coverage: 90%. The interval has one global residual radius.



45.2%

coverage for Marginal rows

Three notebooks cover the full workflow

Run the notebooks in order

```
EDA → Preprocessing → Model Application
```

Explore and audit the data, fit training-only transforms, then train and compare the models.

The model notebook reloads saved weights and predicts new measurements. Charts are exported to figures/.

Results describe index emulation, with explicit limitations.

What the results establish

What we verified

A documented target audit
A fair tree-versus-MLP comparison
Measured explanations and coverage
A reproducible inference workflow

Research limits

Target depends on the same inputs
Prior imputation and unit questions
Unequal country representation
One seed and bounded tuning

A stronger study needs independent outcomes or a defined reduced sensor panel.

Conclusion and references

XGBoost is the stronger individual model.
The blend gives a small RMSE improvement.

When every formula input is available, the transparent source calculation is the best way to reproduce this exact index.

Karim et al. (2025), Surface Water Quality Dataset, Figshare v2
CCME (2017), Water Quality Index User Manual
Chen & Guestrin (2016), XGBoost, KDD
Goodfellow et al. (2016), Deep Learning, chapter 6
Angelopoulos & Bates (2021), Conformal Prediction, arXiv:2107.07511

github.com/hypnoastic/Water_Quality_Prediction